

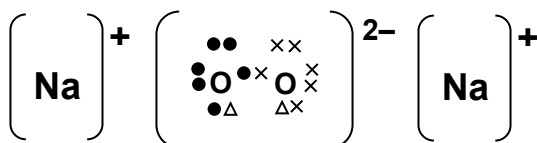
Suggested Answers for 2018 H2 Chemistry Preliminary Examination Paper 3

1 (a) (i) H_2O_2 :



[1m]

Na_2O_2 :



[1m]

Accept:

1) **2** $[\text{Na}]^+$

Reject:

1) $[\text{Na}]_2^+$

2) electrons are drawn around Na^+

- (ii) Sodium peroxide has a giant ionic structure while hydrogen peroxide has a simple molecular structure.

More energy is needed to overcome the **strong ionic bonds between Na^+ and O_2^{2-} /oppositely charged ions** (✓)

compared to the **weak intermolecular force in H_2O_2 / hydrogen bonds between H_2O_2 molecules**. (✓)

Hence, sodium peroxide has a higher boiling point than hydrogen peroxide.

Both sodium peroxide and sodium oxide have a giant ionic structure.

While O_2^{2-} and O^{2-} have the same ionic charge, **O_2^{2-} has a larger radius/ bigger size than O^{2-}** (✓).

Less energy is needed to overcome the **weaker ionic bonds between Na^+ and O_2^{2-} than that between Na^+ and O^{2-}** . (✓)

Hence, sodium peroxide has a lower boiling point than sodium oxide.

4(✓) : [2m] ; 2-3(✓) : [1m]

Minus 1m if no comparison of the strength of bonds is made.

(b) (i) $\text{SrO}_2 \rightarrow \text{SrO} + \frac{1}{2} \text{O}_2$ or $2\text{SrO}_2 \rightarrow 2\text{SrO} + \text{O}_2$ **[1m]**

- (ii) Down the Group, **decomposition temperature increases**. **[1m]**

Down the Group,

① **radius of M^{2+} /cation increases**

② **charge density of M^{2+} decreases**

} **[1m]**

③ **polarising power of M^{2+} decreases / polarisation of O_2^{2-} anion occurs to smaller extent**

④ **less weakening of O–O bond**

⑤ MO_2 becomes **thermally more stable**, higher temperature is needed to decompose.

} **[1m]**

1 (c) (i) $pK_a(1)$ values are all smaller than the pK_a of ethanoic acid

[1m] smaller pK_a implies larger K_a , indicating dicarboxylic acids are more acidic/ stronger acids than ethanoic acid

[1m] $-\text{CO}_2\text{H}$ group is electron-withdrawing, disperse the negative charge and stabilise monoanion/ $\text{HO}_2\text{C}(\text{CH}_2)_n\text{CO}_2^-$

Or monoanion/ $\text{HO}_2\text{C}(\text{CH}_2)_n\text{CO}_2^-$ is stabilised by hydrogen bonding (especially for malonic acid when $n=1$)

$pK_a(1)$ values become larger as n increases

[1m] electron-withdrawing $-\text{CO}_2\text{H}$ group is further away from $-\text{CO}_2^-$ / ionising $-\text{CO}_2\text{H}$ / the other $-\text{CO}_2\text{H}$ group

Or intervening additional electron-donating alkyl groups destabilise anion

(ii) larger pK_a implies smaller K_a with $\text{HO}_2\text{C}(\text{CH}_2)_n\text{CO}_2^-$ being a weaker acid than $\text{CH}_3\text{CO}_2\text{H}$; it is more difficult to remove a positively charged H^+ from a negatively charged species/ monoanion (electrostatically not favourable) [1m]

$$\begin{aligned} \text{(iii)} \quad [\text{H}^+] &= \sqrt{K_a \times c} \\ &= \sqrt{10^{-2.83} \times 0.10} \\ &= 0.0122 \text{ mol dm}^{-3} \end{aligned}$$

$$\text{pH} = -\lg [\text{H}^+] = 1.92 \quad [1\text{m}]$$

(iv)

The following should be clearly marked and labelled:

(✓) Both axes are labelled with units stated for x-axis.

(✓) Curve starts at pH 1.92 (ecf)

(✓) First maximum buffering capacity at 5 cm^3 and pH 2.83.

(✓) Second maximum buffering capacity at 15 cm^3 and pH 5.69.

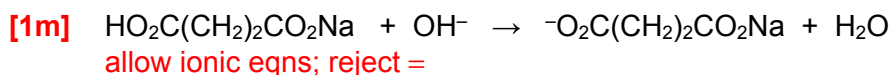
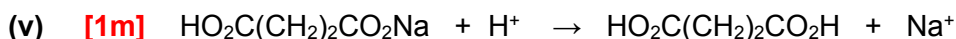
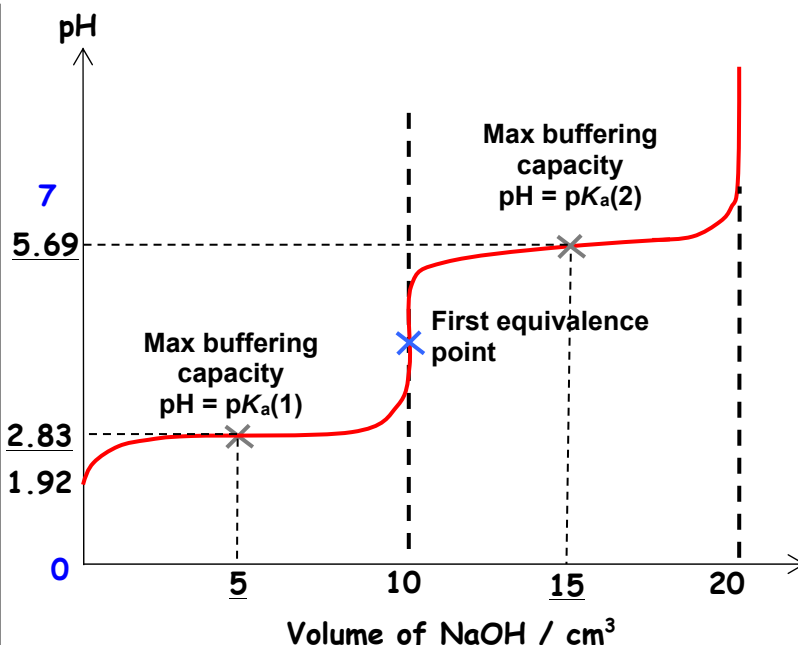
(✓) First equivalence point at 10 cm^3 and graph ends at 20 cm^3

(✓) 2 points of inflection

6(✓) – [3m]

4–5(✓) – [2m]

2–3(✓) – [1m]



$$\begin{aligned} \text{pH} &= \text{pK}_a + \lg \frac{[\text{salt}]}{[\text{acid}]} = \text{pK}_a + \lg \frac{n(\text{salt})}{n(\text{acid})} \\ &= 5.61 + \lg \left(\frac{\frac{50}{1000} \times 0.3}{\frac{100}{1000} \times 0.5} \right) \\ &= \mathbf{5.09} \end{aligned}$$

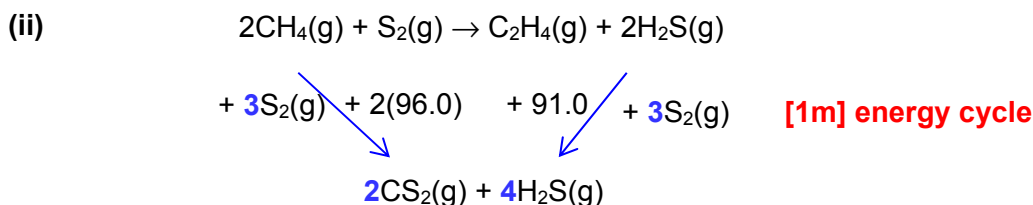
[1m] pH calculation (ecf from wrong pK_a)

- 2 (a) Inter-electronic repulsion between paired 3p electrons makes it easier to remove one of the paired 3p electrons than to remove the unpaired 3p electron from phosphorus. [1m]

- (b) (i) **[1m]** Either one of the following:

Very strong C–C and C–H bonds / high bond energies

Non-polar / C and H have similar electronegativities



$$\Delta H \text{ for reaction 1} = +2(96.0) - 91.0 = \mathbf{+101 \text{ kJ mol}^{-1}} \text{ [1m]}$$

- (c) (i)** $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) = 2\text{SO}_3(\text{g})$

initial	partial	pressure	100	100	0	
/kPa						
			$-0.98(100)$		$-\frac{1}{2}(98)$	
			↙	↘		
eqm	partial	pressure	2 (✓)	51 (✓)	98 (✓)	
/kPa						
						2 (✓): [1m]
						3(✓): [2m]

$$K_P = \frac{(P_{\text{SO}_3})^2}{(P_{\text{SO}_2})^2 (P_{\text{O}_2})} = \frac{(98)^2}{(2)^2 (51)} = 47.1 \text{ kPa}^{-1} \text{ [1m]}$$

- (ii) Since K_p at 25°C is much larger than 1, the position of equilibrium in $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) = 2\text{SO}_3(\text{g})$ lies very much to the right [1m], so the forward reaction is spontaneous and ΔG° is negative. [1m]

As K_p decreases with increasing temperature, it implies that the **position of equilibrium shifts to the left** [1m] with increasing temperature to absorb some heat. Hence, the backward reaction is endothermic and the **forward reaction is exothermic**, so ΔH has a negative sign. [1m]

2 (d) (i) **SO₂** is less ideal.

Either:

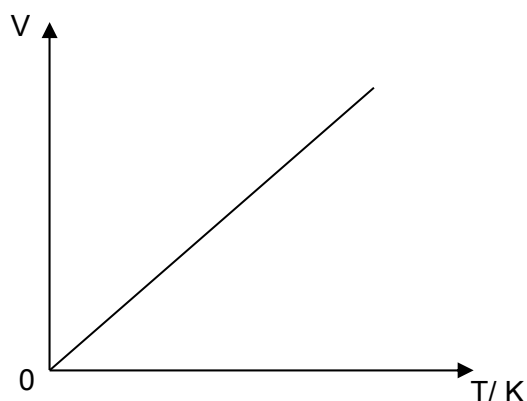
SO₂ has stronger intermolecular forces of attraction between its molecules as it is a **polar molecule / has greater number of electrons per molecule**.

OR

SO₂ is a larger molecule than O₂, so the **volume of SO₂ molecules is less negligible compared to the volume of the container/gas**.

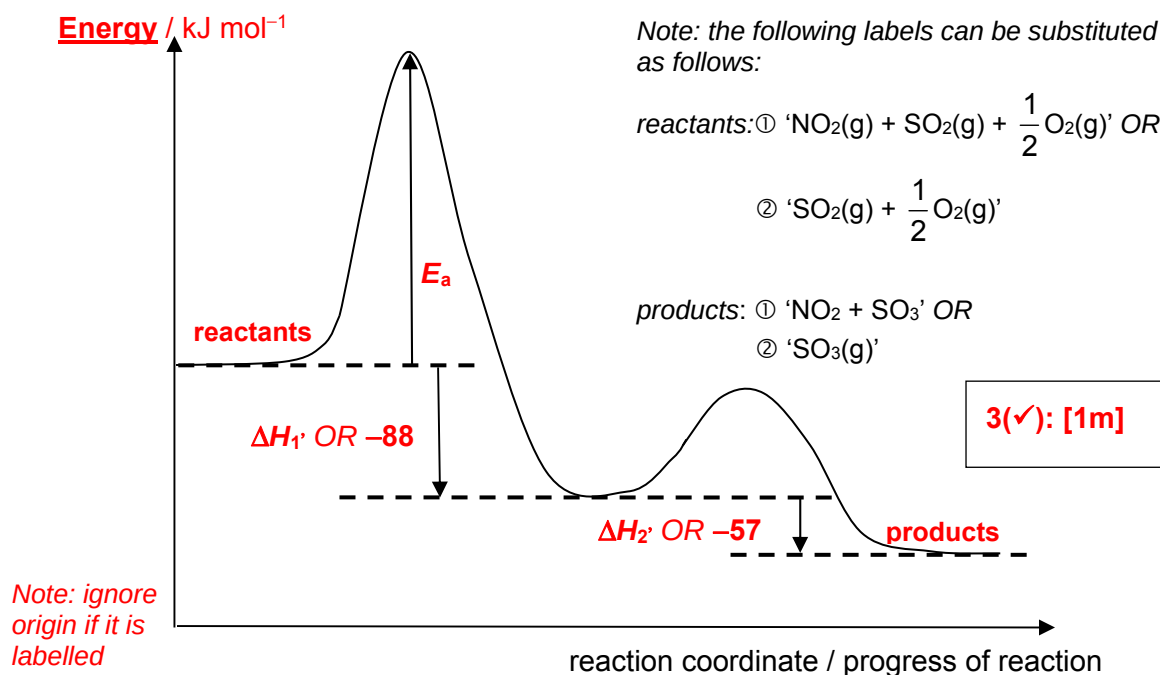
[1m]

(ii)



[1m]

(e)



(✓): 2 humps.

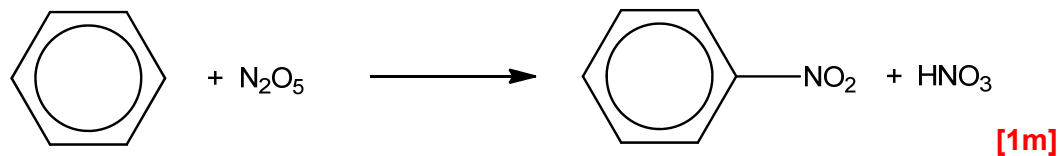
(✓): label 'reactants', 'products' OR using appropriate formulae (with balanced species)

(✓): label y-axis. Ignore x-axis label. Reject if reactants are labelled at '0' level

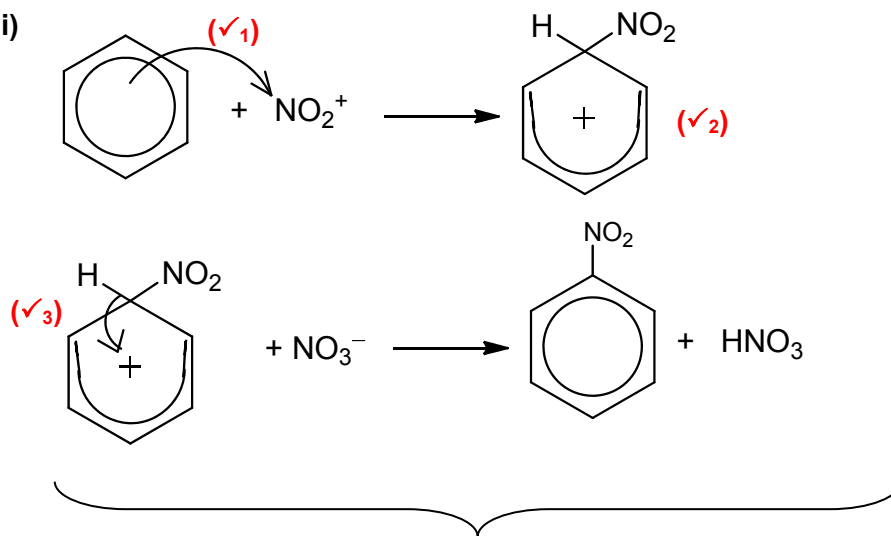
[1m]: Both ΔH₁ and ΔH₂ shown. Ignore direction of arrows.

[1m]: correct labelling of E_a, with E_{a1} > E_{a2}. Ignore direction of arrows.

2 (f) (i)



(ii)



(✓₁) – full arrow from π -electron ring of benzene to the N atom of NO_2^+ .

(✓₂) – correct arenium ion with delocalisation of positive charge over the other 5 carbons.

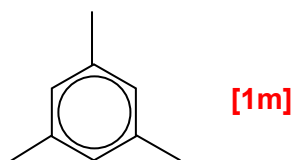
(✓₃) – full arrow from C–H bond to the (+)ve charge of arenium ion

(✓₄) – correct products formed with **balanced equation** with HNO_3 as the other product

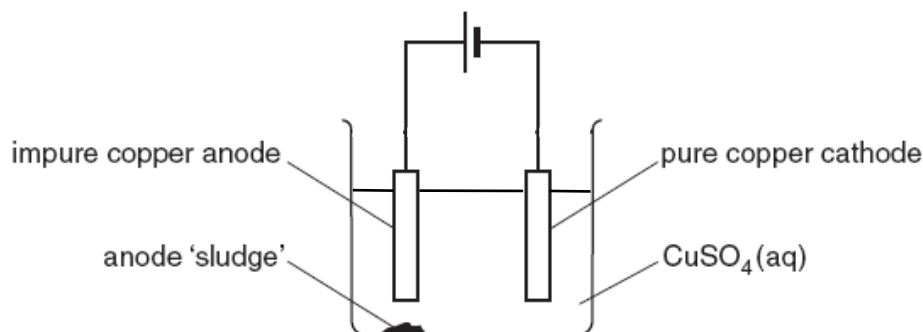
Note: Labelling of “slow/fast” step is **not** required.

4(✓): [2m]; 2–3(✓): [1m]

(iii) Structure of H:



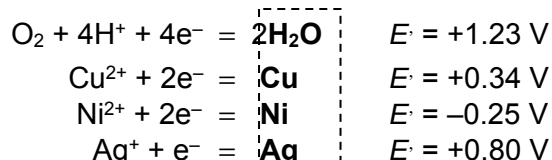
- 3 (a) (i) Cu: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$ [1m]
 (not $3d^9 4s^2$ due to the extra stability of $3d^{10}$ and the similar energies of 3d and 4s electron)
- (ii) Copper is a d-block element that forms one or more stable ions with incompletely filled d-orbitals. [1m]
- (iii) Copper has high density/ is too heavy for overhead use. [1m]
- (b) (i) S is oxidised from -2 to +4 [1m]
- (ii) 2 CuFeS₂ + 2 SiO₂ + 5 O₂ → 2 Cu + 2 FeSiO₃ + 4 SO₂ [1m]
- (iii)



[1m] these labels on the diagram: **impure copper, pure copper, CuSO₄(aq)**

[1m] battery with 2 electrodes dipped in a common electrolyte;
 battery must show the correct polarity with respect to the impure copper.

- (iv) species present:
 Cu²⁺(aq), SO₄²⁻(aq), H₂O(l), impure Cu (anode) and impurities such as Ag, Ni



Note:

The species is more easily oxidised when the E° is more negative (or less positive).

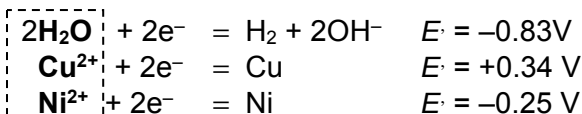
[1m] quote E° values for Cu²⁺/Cu, Ni²⁺/Ni and Ag⁺/Ag

At the anode,

Cu(s) is oxidised to Cu²⁺(aq) in preference over H₂O as E°(Cu²⁺/Cu) is more negative than E°(O₂/H₂O).

Ni(s) is also oxidised to Ni²⁺(aq) (which goes into the solution) as E°(Ni²⁺/Ni) is more negative than E°(Cu²⁺/Cu).

Ag(s) is not oxidised to Ag⁺ as E°(Ag⁺/Ag) is more positive than E°(Cu²⁺/Cu).
 ∴ Ag(s) is collected as "anode sludge".



Note:

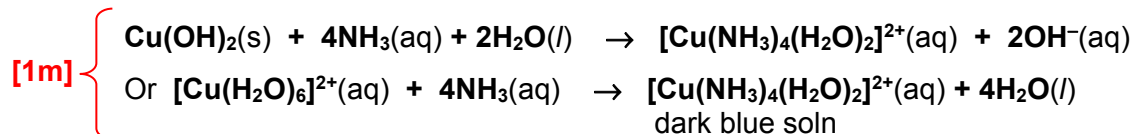
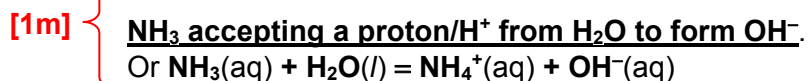
The species is more easily reduced when the E° is more positive (or less negative).

At the cathode,

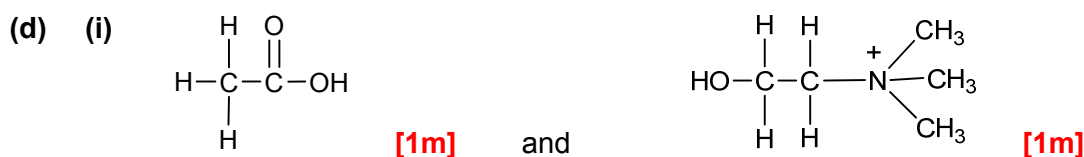
Cu²⁺(aq) is reduced to Cu(s) in preference over H₂O as E°(Cu²⁺/Cu) is more positive than E°(H₂O/H₂).

Ni²⁺(aq) is not reduced to Ni(s) as E°(Ni²⁺/Ni) is more negative than E°(Cu²⁺/Cu).
 ∴ Ni²⁺(aq) remains in the solution.

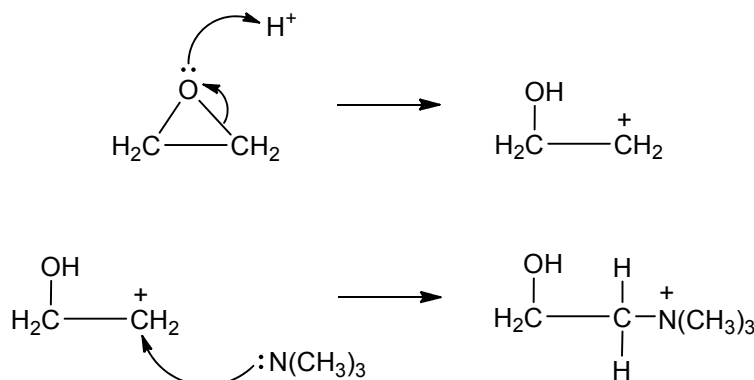
3 (c) **Bronsted-Lowry base** is a **proton/H⁺ acceptor** as illustrated by



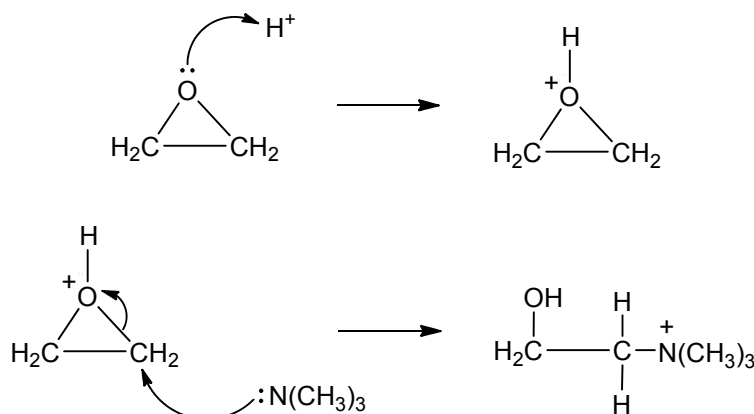
When added in excess, NH₃ acts as a Lewis base where N in NH₃ ligand **donates a lone pair** of electrons **to Cu²⁺**, forming strong dative bonds with Cu²⁺ to give the stable [Cu(NH₃)₄(H₂O)₂]²⁺ complex. [1m]



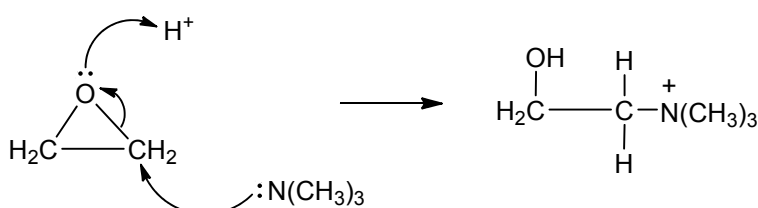
(ii) **Option 1:**



Option 2:

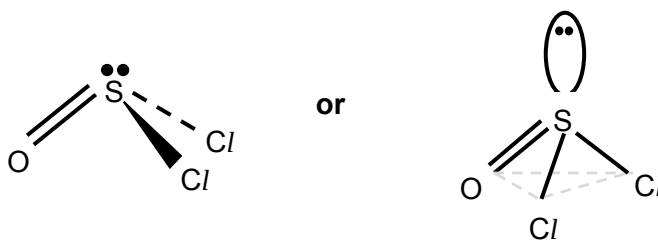


Option 3:



[1m] for each curly arrow with the associated lone pair and any positive charge

4 (a) (i)



[1m] illustrate trigonal pyramidal shape.

(ii) COCl₂ will have a larger bond angle. [1m] bonus

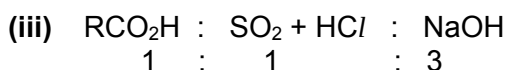
COCl₂ has lesser number/ 3 sets of electron pairs (or state 3 bp, 0 lp) which can be further apart to minimise repulsion compared to SOCl₂ that has 4 electron pairs (or state 3 bp, 1 lp). [1m]

(b) (i) NaOH + HCl → NaCl + H₂O [1m]2 NaOH + SO₂ → Na₂SO₃ + H₂O [1m]

Allow if 2nd eqn is split into 2 eqns: SO₂ + H₂O → H₂SO₃
2NaOH + H₂SO₃ → Na₂SO₃ + 2H₂O

Reject: NaOH + SO₂ → NaHSO₃ for NaOH is added in excess

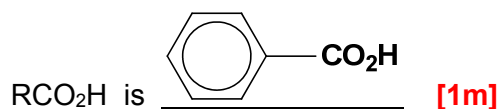
$$\begin{aligned} \text{(ii)} \quad n(\text{NaOH}) \text{ reacted} &= \frac{60.0 - 10.8}{1000} \times 0.500 \\ &= \underline{0.0246} \text{ mol} \quad [1\text{m}] \end{aligned}$$



$$n(\text{RCO}_2\text{H}) = \frac{0.0246}{3} = \underline{0.00820} \text{ mol} \quad [1\text{m}]$$

$$\text{(iv)} \quad M_r(\text{RCO}_2\text{H}) = \frac{1.00}{0.0082} = \underline{122} \quad [1\text{m}] \text{ no units, 3 s.f., ecf}$$

(v) RCO₂H: mass units of -CO₂H = 45
mass units of -R = 122 - 45 = 77
estimated number of C in -R = 77/12 = 6.4 (⇒ R contains 6 carbon atoms)

(c) (i) RCONH₂ [1m](ii) Step 2: NH₃(g) [1m] reject (aq)Step 3: LiAlH₄, dry ether [1m] reject (aq); ignore 'heat'

(iii) Step 2: condensation/ nucleophilic acyl substitution } allow nucleophilic substitution
Step 3: reduction } [1m] each, bonus

4 (d) (i)

Observation	Type of Reaction	Deduction
Angelic acid + $\text{H}_2 \rightarrow \text{T}$ $\text{C}_5\text{H}_8\text{O}_2 \quad \quad \quad \text{C}_5\text{H}_{10}\text{O}_2$	<u>reduction</u> (✓)	<u>Angelic acid</u> is an <u>alkene</u> . (✓)
Angelic acid + $\text{KMnO}_4 \rightarrow \text{U}$ and V	<u>Oxidative cleavage of C=C</u> <u>/oxidation</u> (✓)	<u>U</u> and <u>V</u> are <u>carboxylic acids</u> . (✓)
$\text{U} + \text{I}_2/\text{OH}^- \rightarrow$ yellow ppt	<u>Iodoform test/oxidation</u> (✓)	<u>U</u> has the structure $\text{CH}_3-\text{C}=\text{O}$ (✓) $\quad \quad \quad $

5-6(✓): [2m]; 3-4(✓): [1m]

<u>angelic acid</u>	<u>T</u>	<u>U</u>	<u>V</u>

[1m] for each structure

- (ii) angelic acid : cis-trans isomerism [1m]
 compound T : enantiomerism [1m]

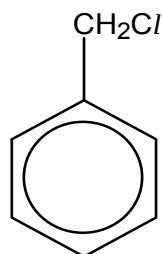
5 (a) (i) SiCl_4 reacts with water / hydrolyses completely in water to give a strongly acidic solution. [1m] **bonus**

A white solid/ppt of SiO_2 will be observed/ white (or steamy) fumes of HCl will be observed. [1m]

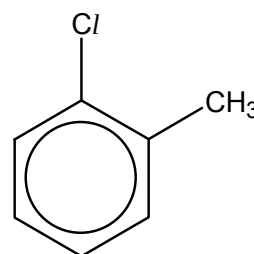


- (ii) Unlike silicon, carbon does not have energetically accessible empty 3d orbitals to accept a lone pair of electrons from water molecules. [1m]

(b)

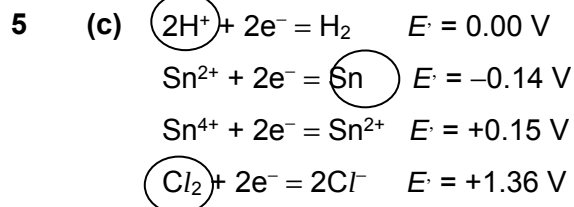


A

B (-CH₃ and Cl groups can be in any position)

[1m] for both structures of A and B

For compound B, delocalisation of the lone pair of electrons on Cl into the π -electron cloud of benzene ring imparts double bond character to the C-Cl, strengthening the C-Cl bond. Hence, B does not react with NaOH(aq) . [1m]



Reaction between H^+ and Sn:

$$E_{\text{cell}}^\circ = 0.00 - (-0.14) = +0.14 \text{ V} > 0 \text{ (reaction is energetically feasible) [1m]}$$

Reaction between H^+ and Sn^{2+} :

$$E_{\text{cell}}^\circ = 0.00 - (+0.15) = -0.15 \text{ V} < 0$$

Oxidation of Sn^{2+} to Sn^{4+} by H^+ is NOT energetically feasible.

Reaction between Cl_2 and Sn:

$$E_{\text{cell}}^\circ = 1.36 - (-0.14) = +1.50 \text{ V} > 0$$

Oxidation of Sn to Sn^{2+} by Cl_2 is energetically feasible. [1m]

Reaction between Cl_2 and Sn^{2+} :

$$E_{\text{cell}}^\circ = 1.36 - (+0.15) = +1.21 \text{ V} > 0$$

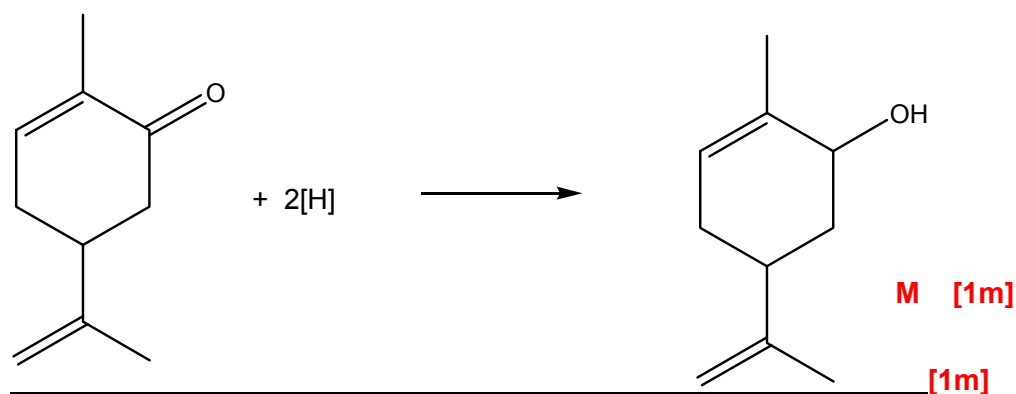
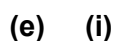
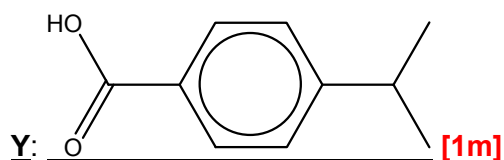
Oxidation of Sn^{2+} to Sn^{4+} by Cl_2 is energetically feasible. [1m] bonus

Award max [1m] if three E_{cell}° values are calculated without any comment.



(ii) Step 2: alkaline $\text{I}_2(\text{aq})$, heat, followed by $\text{H}^+(\text{aq})$ [1m]

Step 3: LiAlH_4 , dry ether [1m]



5 (e) (ii)

Information from question	Type of reaction	Deduction
N reacts with hot concentrated KMnO_4 to give P and Q	<u>oxidation</u> (✓)	<u>C=C bond in N cleaves</u> (✓)
N , P , Q react with 2,4-DNPH	<u>condensation</u> (✓)	<u>N, P and Q contains carbonyl group / contains either aldehyde or ketone groups</u> (✓)
Only N gives silver mirror with Tollens' reagent	<u>oxidation</u> (✓)	<u>N contains aliphatic aldehyde group</u> (✓) <u>P and Q contain ketone group</u> (✓)
One mole of Q reacts with two moles of $\text{NaHCO}_3(\text{aq})$	<u>acid-carbonate reaction / acid – hydrogen carbonate reaction</u> (✓)	<u>Q contains 2 $-\text{CO}_2\text{H}$ groups</u> (✓)

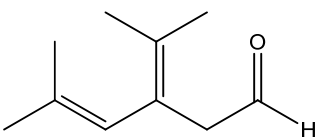
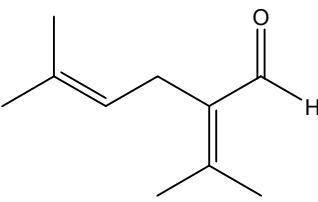
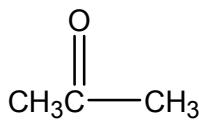
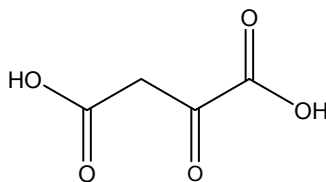
8-9(✓): [4m]

6-7(✓): [3m]

4-5(✓): [2m]

2-3(✓): [1m]

[1m] for each correct structure

 or 		
N	P	Q